Complete EFX for Four Additive Agents and Eight Goods

A Hybrid Lean and CPC Computer-Assisted Proof

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Abstract

We prove that every instance of four agents and eight indivisible goods with nonnegative additive valuations admits a complete allocation that is envy-free up to any good (EFX). This closes the first previously open four-agent additive parameter cell beyond the general $m \leq n + 3$ frontier.

The proof is hybrid. Lean verifies a structural reduction from a hypothetical counterexample to eight precisely specified terminal ranking cells. Each terminal cell is translated into quantifier-free rational linear arithmetic. An independent parser checks every SMT assertion against a separately implemented mathematical specification, while the correspondence between that specification and the relevant Lean predicates is an explicitly identified human-audited trust boundary. For each formula, cvc5 emits a CPC proof of unsatisfiability and the separate Ethos checker returns `correct`. The eight proofs contain 13,812,082 proof-step lines and occupy 1.668 GB. A second encoding over all 40,824 labeled onto allocations independently closes all eight cells. An exact comparison additionally finds that all 545,360 SMT assertions occur as the 545,360 assumptions of the CPC proofs, with zero mismatches.

We state the complete conditional Lean interface, including a kernel-checked perturbation argument from nonnegative to positive weights, explain the formal-to-computational boundary, and provide a content-addressed reproduction package. We also extract a 130-partition deletion-minimal core for one terminal cell, with its own independently checked CPC proof, and isolate an invariant 191-partition statement as a target for a concise human proof. The result is a bounded computer-assisted mathematical theorem, not a resolution of additive EFX in general or a claim that the entire proof is internal to Lean's kernel.

Keywords. Fair division, indivisible goods, EFX, additive valuations, computer-assisted proof, Lean, SMT, CPC.

1 Introduction

Let $\mathcal{N}$ be a finite set of agents and $\mathcal{M}$ a finite set of indivisible goods. Agent i has an additive valuation

$$v_i(S) = \sum_{g \in S} w_{ig}, \qquad w_{ig} \geq 0.$$

A complete allocation $X = (X_i)_{i \in \mathcal{N}}$ partitions all goods. It is *envy-free up to any good* (EFX) if, for every pair $i \neq j$ and every good $g \in X_j$,

$$v_i(X_i) \geq v_i(X_j \setminus \{g\}).$$

Whether complete EFX allocations always exist for additive valuations is a central open problem in discrete fair division.

Complete EFX is known for three additive agents [1]. Mahara proved complete EFX for general monotone valuations when $m \leq n + 3$ [2]. Hence four agents and seven goods are covered, whereas four agents and eight goods form the first previously open four-agent additive cell beyond that few-goods boundary. For four additive agents, an EFX allocation leaving at most one good unallocated is known [3], and complete EF2X is known [4]; neither implies complete EFX.

Our main theorem settles this fixed cell.

Theorem 1.1 (Main theorem)**.** *Every nonnegative additive valuation instance with four agents and eight goods admits a complete EFX allocation.*

The finite-search reduction is carried out for strictly positive weights: positivity implies that every complete EFX allocation has nonempty bundles. This does not restrict the theorem. The set of weight matrices with no complete EFX allocation is open, because it is the complement of a finite union of closed polyhedra, one for each complete allocation. Thus a hypothetical counterexample with zero-valued goods remains a counterexample after adding the same sufficiently small $\varepsilon > 0$ to every item weight. Lean verifies this closure step.

1.1 Proof architecture

The proof combines four layers:

1. published complete-EFX inputs for four agents and seven goods and for three additive agents;
2. a Lean-checked mathematical reduction to eight terminal ranking cells;
3. eight independently generated and parser-audited exact linear-arithmetic formulas; and
4. eight cvc5 CPC proofs accepted by the Ethos checker.

The division of labor is deliberate. Solver output does not determine the mathematical reduction. Conversely, Lean does not replay 13.8 million CPC steps. The complete result is a hybrid theorem whose trusted components are stated explicitly in Section 10.

1.2 Contributions

Besides Theorem 1.1, the development provides:

- a formal perturbation lemma extending the positive terminal proof to all nonnegative additive valuations;
- a reverse-parking criterion and a terminating threshold-cycle normal form for EFX allocations;
- a favorite-small-bundle theorem, including a constructive EFX result whenever the favorite map is injective and $n \leq m \leq 2n$;
- formal universal-minimum and peelable residual reductions;
- exact canonical coverage and valid stabilizer reductions;
- a partition-to-perfect-matching formulation covering all 1,701 unlabeled four-bundle partitions;
- independent terminal and direct-allocation encodings;
- exact CPC proof objects for every terminal cell; and
- a deletion-minimal 130-partition explanatory core.

1.3 What the result does and does not settle

Theorem 1.1 is a bounded existence theorem. It does not prove EFX for four agents and an arbitrary number of goods, nor does it resolve the general additive EFX conjecture. Its significance is that it crosses the first previously open four-agent parameter cell at which the existing general few-goods theorem stops.

2 Related work and parameter frontier

For additive valuations, complete EFX is known for three agents [1]. For any number of agents, Mahara proved existence for $m \leq n + 3$ under general monotone valuations [2]. The four-agent partial result of Berger et al. leaves at most one good unallocated [3]. Ashuri, Gkatzelis, and Sgouritsa prove EF2X for four agents and explicitly distinguish that relaxation from the open EFX problem [4].

Neoh and Teh study counting and variants of EFX, recording the same $m \leq n+3$ few-goods frontier [5]. Brânzei reports strong empirical performance of a local-search heuristic on additive instances but does not obtain unrestricted existence [6]. Recent positive results for hypergraph valuations and for two types of goods remain structured subclasses of the general additive setting [8, 9].

Akrami et al. give nonexistence examples from $m \geq n + 5$ for monotone and submodular valuations and formally verify their SAT foundations in Lean [7]. They also prove complete EFX existence for three agents and seven goods under monotone valuations. That positive result is likewise an $m = n + 4$ cell and uses certified finite search, so the broad proof strategy has direct recent precedent. The present result is its four-agent additive analogue, using exact linear arithmetic and CPC/Ethos rather than SAT/DRAT. Their counterexamples are nonadditive and begin one good later. The current landscape is therefore

$$\text{general monotone existence through } m = n + 3,$$

$$\text{monotone existence at } (3, 7),$$

$$\boxed{\text{nonnegative additive existence at } (n, m) = (4, 8)},$$

$$\text{nonadditive failure from } m = n + 5.$$

Akrami et al. explicitly leave the $m = n + 4$ and four-or-more-agent additive frontiers open. Our theorem closes one fixed subcase of those questions; it is not a general solution and does not claim methodological priority.

3 Model and elementary structure

Definition 3.1 (Complete allocation)**.** An allocation is a function assigning every good to exactly one agent. Its bundles X_i are pairwise disjoint and have union $\mathcal{M}$.

Definition 3.2 (EFX)**.** A complete allocation X is EFX if

$$v_i(X_i) \geq v_i(X_j \setminus \{g\})$$

for every distinct pair i, j and every $g \in X_j$.

For an agent i and bundle S, define the one-good-removal threshold

$$\tau_i(S) = \max_{g \in S} v_i(S \setminus \{g\}), \qquad \tau_i(\varnothing) = 0.$$

For additive valuations,

$$\tau_i(S) = v_i(S) - \min_{g \in S} v_i(g)$$

when S is nonempty. An allocation is EFX exactly when

$$v_i(X_i) \geq \tau_i(X_j) \quad \text{for all } i, j.$$

3.1 Reverse parking

Suppose X is EFX and $f \in X_h$. Removing f only makes the owner comparison for h harder; every other target bundle either stays fixed or shrinks. Consequently, f is removable from h precisely when

$$v_h(X_h) - v_h(f) \geq \max_{j \neq h} \tau_h(X_j).$$

Choosing a least-valued good in X_h gives:

Lemma 3.3 (Reverse parking)**.** *An owner h has a removable good if and only if*

$$\tau_h(X_h) \geq \max_{j \neq h} \tau_h(X_j).$$

If no owner satisfies this inequality, choose for every owner h a strict threshold-maximizing target $c(h)$. The resulting functional graph contains a directed cycle. Rotating bundles along that cycle preserves EFX and strictly increases the finite potential

$$\Phi(X) = \sum_h \tau_h(X_h).$$

Theorem 3.4 (Threshold-cycle normal form)**.** *Every complete EFX allocation with nonempty bundles can be transformed by EFX-preserving bundle rotations into one having an owner-removable good.*

The formal theorem is generic over finite agents and goods and nonnegative additive valuations. In the positive 4×8 setting, an empty bundle would force every other bundle to have cardinality at most one, contradicting completeness; hence all bundles are nonempty.

3.2 From nonnegative to positive weights

Let $\mathcal{A}$ be the finite set of complete allocations. For a fixed $X \in \mathcal{A}$, the set of weights for which X is EFX is an intersection of finitely many closed linear half-spaces. Hence

$$\{w : w \text{ admits a complete EFX allocation}\} = \bigcup_{X \in \mathcal{A}} \{w : X \text{ is EFX}\}$$

is closed, and the set of counterexamples is open.

Lemma 3.5 (Positive approximation)**.** *If a nonnegative additive 4×8 instance has no complete EFX allocation, then some strictly positive additive 4×8 instance has no complete EFX allocation.*

Proof. For $\varepsilon \in \mathbb{R}$, set $w_{ig}^{\varepsilon} = w_{ig} + \varepsilon$. The path $\varepsilon \mapsto w^{\varepsilon}$ is continuous and passes through w at $\varepsilon = 0$. Openness of the no-EFX set gives a radius $\delta > 0$ such that w^{ε} still has no EFX allocation whenever $|\varepsilon| < \delta$. Taking $0 < \varepsilon < \delta$ makes every coordinate strictly positive. □

Lean proves the closedness of each fixed-allocation EFX set, the finite-union closedness of the existence set, continuity of the uniform perturbation, and Lemma 3.5. The remainder of the reduction may therefore work with positive weights without narrowing Theorem 1.1.

3.3 Normalization and generic strictness

Multiplying one agent's valuation row by a positive scalar preserves every EFX comparison. We normalize each row so that

$$\sum_{g \in \mathcal{M}} w_{ig} = 1.$$

There are finitely many complete allocations. If a normalized instance has no EFX allocation, choose a strict failed inequality for each allocation. These finitely many failures persist in an open neighborhood. A sufficiently small perturbation within the row-sum hyperplanes preserves positivity and nonexistence while avoiding both coordinate-equality hyperplanes within each row and equality of any two full valuation rows.

Lemma 3.6 (Generic normalized counterexample)**.** *If a strictly positive additive 4×8 instance has no EFX allocation, then there is a row-normalized counterexample in which every row has pairwise distinct item values and the four valuation rows are pairwise distinct.*

Lean proves the simultaneous perturbation, rather than using genericity as an informal without-loss-of-generality step.

3.4 Favorite small bundles

Lemma 3.7 (Favorite-small-bundle lemma)**.** *Let X be a complete allocation in a positive additive instance. If every agent owns one of its favorite goods and every bundle has size at most two, then X is EFX.*

Proof. After removing any good from a target bundle, at most one good remains. The observer values that good no more than its own favorite, which is contained in its own bundle. □

The formal result is constructive. If the favorite map is injective and $|\mathcal{M}| \leq 2|\mathcal{N}|$, it assigns every agent its favorite and distributes at most one residual good per agent. Thus the injective-favorite case is closed at 4×8.

4 Reduction to eight terminal cells

Assume for contradiction that a nonnegative additive 4×8 instance has no complete EFX allocation. First apply Lemma 3.5, then normalize and perturb the resulting positive instance using Lemma 3.6.

4.1 Residual cuts from smaller theorems

Two regions reduce to published smaller cases.

Universal minimum. If some good is minimum for every agent, delete it and apply Mahara's complete 4×7 theorem. The threshold-cycle normal form moves the smaller allocation to an appropriate unenvied source; inserting the deleted minimum good there preserves every EFX comparison.

Peelable agent-good pair. If an agent-good pair satisfies the formal peelability inequalities, delete that agent and that good, apply the complete three-agent theorem to the remaining seven goods, and reinsert the pair.

The peelable predicate excludes the deleted agent as an observer in the smaller instance but retains every active-observer comparison required after reinsertion. This observer scope was corrected during adversarial specification review and is now part of the checked formal statement.

Proposition 4.1 (Residual exclusion)**.** *A normalized generic counterexample has neither a universally minimum good nor a peelable agent-good pair.*

4.2 Favorite multiplicities

By Lemma 3.7, the four favorite goods cannot be distinct. With strict rankings, every noninjective favorite multiplicity is, up to agent and good relabeling,

$$2+1+1, \qquad 3+1, \qquad 2+2, \qquad 4.$$

Canonical favorite vectors for the first three are

$$(0,0,1,2), \qquad (0,0,0,1), \qquad (0,0,1,1).$$

If all four agents favor good 0, refine by the multiplicities of their second favorites:

$$1+1+1+1, \qquad 4, \qquad 3+1, \qquad 2+2, \qquad 2+1+1.$$

This produces exactly eight terminal ranking cells.

4.3 Valid stabilizers

Once a ranking prefix is fixed, only symmetries preserving that prefix remain valid. Agents with identical prescribed prefixes and goods absent from every prefix can be sorted lexicographically. Lean proves that every orbit has a representative satisfying the corresponding stabilizer inequalities. No symmetry exchanging inequivalent favorite or second-favorite labels is used.

Theorem 4.2 (Eight-cell reduction)**.** *Assume the standard four-agent/seven-good and three-agent additive EFX inputs. Every positive additive* 4×8 *counterexample yields a normalized valuation matrix satisfying one of the eight terminal cell specifications in Table 1.*

Cell	Ranking data	Stabilizer
favorite-211	$(0,0,1,2)$	none
favorite-31	$(0,0,0,1)$	favorite preserving
favorite-22	$(0,0,1,1)$	favorite preserving
second-1111	$(1,2,3,4)$	prefix preserving
second-4	$(1,1,1,1)$	prefix preserving
second-31	$(1,1,1,2)$	prefix preserving
second-22	$(1,1,2,2)$	prefix preserving
second-211	$(1,1,2,3)$	prefix preserving

Table 1: The eight terminal ranking cells. The last five share favorite vector $(0,0,0,0)$.

5 Partitions and perfect matchings

There are exactly

$$S(8,4) = 1,701$$

unlabeled partitions of eight goods into four nonempty bundles. The Lean development verifies the restricted-growth enumeration and proves that every onto labeled allocation is obtained from one enumerated partition by a permutation of the bundles.

Fix a partition $\mathcal{B} = (B_0, B_1, B_2, B_3)$. Construct a bipartite graph between agents and bundles. Put edge iB_b when

$$v_i(B_b) \geq \max_{c \neq b} \tau_i(B_c).$$

Proposition 5.1 (Matching equivalence)**.** *The partition $\mathcal{B}$ supports a complete EFX allocation if and only if its agent-bundle feasibility graph has a perfect matching.*

Proof. A perfect matching assigns each bundle to one distinct agent. The edge condition says that the assigned owner passes every one-good-removal comparison against the three other bundles, hence the allocation is EFX. Conversely, the owner assignment of any complete EFX allocation gives exactly such a matching. □

Therefore a counterexample must obstruct all $4! = 24$ matchings for every one of the 1,701 partitions. This is a finite disjunction of linear comparisons, which can be encoded in exact rational linear arithmetic.

6 Exact terminal formulas

For each cell c, let F_c be a quantifier-free rational linear-arithmetic formula over the 32 item-value variables. It contains:

- 32 strict positivity assertions and four row-normalization equations;
- pairwise row inequalities and the exact favorite or second-favorite order;
- cell-preserving stabilizer inequalities;
- exclusions of universal-minimum and peelable residual regions;
- 27,216 agent-bundle feasibility-edge definitions; and
- 40,824 perfect-matching obstruction assertions.

Proposition 6.1 (Semantic implication)**.** *For every terminal cell c, any valuation matrix satisfying the Lean terminal specification for c satisfies F_c.*

This proposition describes the load-bearing bridge. The current assurance is hybrid:

1. a formula generator independent of the original Z3 encoder emits F_c;
2. a separate parser reconstructs and checks every SMT assertion against a hand-written mathematical specification;
3. the parser independently enumerates all 1,701 partitions and all 24 matching clauses per partition;
4. canonical vectors are checked directly against their Lean source constants; and
5. the remaining correspondence between the hand-written specification and Lean's residual predicates, stabilizers, feasible edge, and matching obstruction has been manually reviewed.

Thus Proposition 6.1 is not presently a single reflected Lean theorem. It is an explicit trusted software-and-review boundary. This is stronger than relying on an unchecked generator, but weaker than kernel-checked extraction from Lean.

7 CPC certificate closure

For each formula F_c, cvc5 emits a CPC proof and Ethos checks the proof under pinned CPC signatures. Table 2 gives the final proof sizes.

Cell	Proof bytes	Proof-step lines
favorite-211	187,295,060	1,890,346
favorite-31	459,896,059	3,327,798
favorite-22	338,399,888	2,702,364
second-1111	21,220,067	249,283
second-4	449,335,513	3,374,635
second-31	115,319,771	1,186,530
second-22	78,871,768	872,213
second-211	18,119,773	208,913
Total	1,668,457,899	13,812,082

Table 2: CPC proof sizes. Ethos returns `correct` for every row.

7.1 Formula-to-proof assumption audit

A valid proof for a different inconsistent formula would not establish unsatisfiability of F_c. To exclude this stale-proof attack, the CPC term definitions were expanded and every proof assumption was compared with the SMT assertion stream. Across all eight cells:

$$545,360 \text{ formula assertions} = 545,360 \text{ proof assumptions},$$

with zero mismatches. The release package includes the executable comparison and a manifest-bound JSON result.

7.2 Direct-allocation cross-check

As non-load-bearing corroboration, a second formulation directly negates every labeled onto allocation. There are

$$4!\, S(8,4) = 40,824$$

such allocations. This formulation does not use the partition feasibility graph. All eight direct formulas are UNSAT: six close in Z3, while the two hardest close in cvc5 after an audited identity-only SMT-LIB syntax normalization.

The direct results increase confidence that no partition-to-matching coding error drives the main certificate, but the six Z3 results do not carry independently checked proof objects and are not part of the load-bearing deduction.

8 Formal composition

Lean proves the structural theorem through the terminal cells. The complete exported interface is:

```
theorem admitsEFX_4x8_of_standard_published_inputs_nonnegative
    (four_agent_seven_good :
      FourAgentSevenGoodStandardInput)
    (three_agent_seven_good :
      ThreeAgentSevenGoodStandardInput)
    (terminal_cells_unsat :
      EightTerminalCellsUnsat)
    (weight : Weight48)
    (nonnegative : forall agent good, 0 <= weight agent good) :
    AdmitsEFX weight
```

Here `EightTerminalCellsUnsat` is the proposition that each of the eight terminal specifications is empty. It is discharged externally by Proposition 6.1 and the CPC/Ethos results, not by a theorem internal to Lean.

The axiom audit of the conditional Lean theorem reports exactly

[`propext`, `Classical.choice`, `Quot.sound`],

with no `sorryAx` and no native-computation axiom in the dependency closure.

Theorem 8.1 (Hybrid closure)**.** *Assume:*

(i) the two published smaller-case EFX theorems;

(ii) the soundness of Proposition 6.1; and

(iii) correctness of Ethos under the pinned CPC signatures.

Then Theorem 1.1 follows from the eight archived CPC proofs.

Proof. The CPC proofs and Ethos verdicts establish $\neg F_c$ for every terminal cell c. By Proposition 6.1, no Lean terminal specification is inhabited, giving `EightTerminalCellsUnsat`. Applying the conditional Lean theorem with the two published inputs yields `AdmitsEFX weight` for every nonnegative 4×8 weight matrix. Internally, the exported theorem first uses Lemma 3.5 to reduce a hypothetical nonnegative counterexample to a positive one. □

9 The 130-partition explanatory core

The favorite-211 cell has favorite vector $(0, 0, 1, 2)$. Starting from all 1,701 partition obstructions, assumption-core extraction produced 200 partitions. Deletion shrinking then produced a 130-partition core.

Bundle-size shape	Initial core	Final core
$3+2+2+1$	98	59
$2+2+2+2$	51	20
$4+2+1+1$	30	30
$3+3+1+1$	20	20
$5+1+1+1$	1	1
Total	200	130

Table 3: Shape counts in the explanatory core.

Every single-partition deletion is satisfiable, so every survivor is indispensable relative to this core. This is deletion minimality, not minimum-cardinality optimality.

The final core has two pieces:

1. 110 partitions in which shared favorite good 0 is a singleton, favorite goods $0, 1, 2$ are separated, and no nonfavorite good is a singleton; and

2. 20 partitions of shape $2 + 2 + 2 + 2$ in which good 0 is paired with one nonfavorite good.

The reduced formula has a 220,777,714-byte CPC proof containing 2,032,963 proof-step lines, accepted by Ethos.

These 130 partitions lie in a simpler invariant family $\mathcal{A}$:

1. all 131 partitions in which good 0 is a singleton, goods $0, 1, 2$ are separated, and no nonfavorite good is a singleton; and

2. all 60 partitions into four pairs in which goods $0, 1, 2$ are separated.

Thus $|\mathcal{A}| = 191$. The certificate proves the following weaker but more symmetric statement, because its 130 obstructions are a subset of $\mathcal{A}$.

Proposition 9.1 (Anchored contested-favorite lemma)**.** *Let w be a positive, row-normalized 4×8 valuation with distinct rows, strict favorite vector $(0, 0, 1, 2)$, no universally minimum good, and no peelable agent-good pair. Then some partition in $\mathcal{A}$ has a feasible perfect matching.*

Proposition 9.1 is currently a certificate corollary, not a human proof. The Lean development defines its exact invariant family and proves that this single statement suffices to eliminate the favorite-211 terminal cell. It is the most concrete analytic replacement target produced by the core analysis.

Exact deletion witnesses have been extracted for the first 34 of the 130 partitions. For each completed deletion, the removed partition is the sole supported partition within the 130-core instance. It need not be unique in the full 1,701-partition universe. Successful removed partitions also typically have several perfect matchings.

The Hall data also rejects a tempting compression. In the resulting 4,386 obstructed feasibility graphs there are 426 distinct graph signatures. Although the full four-agent Hall witness is the most frequent primary witness, an exact set-cover calculation needs ten different nontrivial agent masks to cover the sample, and that size-ten cover is unique. Frequencies of three dominant masks therefore do not support a three-case proof. The promising compression is geometric—prove Proposition 9.1 using the two bundle families and threshold inequalities—rather than a taxonomy of Hall masks.

10 Trust boundary and limitations

The complete deduction has the form

$$\begin{aligned} &\text{two pinned published EFX theorems} \\ &+ \text{ Lean-checked conditional structural reduction} \\ &+ \text{ parser and human audited Lean-to-SMT implication} \\ &+ \text{ eight CPC proofs accepted by Ethos} \\ &\Longrightarrow \text{ nonnegative additive EFX at } (4, 8). \end{aligned}$$

10.1 Trusted components

The remaining trusted components are:

- the mathematical proofs in the two published input papers;

- the Lean kernel, pinned toolchain, mathlib, and pinned dependencies;

- the Ethos checker and pinned CPC signatures;
- the parser implementation; and
- the manually reviewed correspondence between the parser's mathematical specification and the relevant Lean definitions.

10.2 Limitations

- The entire proof is not one theorem checked inside Lean's kernel.
- The two published paper proofs are not re-formalized.
- The theorem treats only four agents and eight goods.
- The explanatory core does not replace the full terminal certification.
- Same-host reproduction does not substitute for independent-machine reproduction.
- Novelty has been searched in current primary literature but requires ordinary expert and peer-review confirmation.

10.3 Independent internal adversarial reviews

Two read-only internal reviewers worked independently. The mathematical review found no material reduction error, conditional on terminal UNSAT. The formal/computational review found no formula or proof mismatch and contributed the 545,360-assumption comparison. Both reviewers independently identified the semantic bridge as the highest-risk boundary. These reviews are evidence of adversarial inspection, not external peer review.

11 Reproducibility and artifact package

The final content-addressed package contains:

- a deterministic source archive containing the Lean project, independent generators, and executable audits;
- the eight exact terminal formulas and CPC proofs;
- pinned CPC signatures and tool acquisition hashes;
- the semantic, published-input, direct-allocation, and formula-to-proof audits;
- the Lean build and axiom-audit transcript;
- the two internal referee reports;
- this submission manuscript; and
- machine-readable manifests and replay instructions.

Every load-bearing entry is hash-bound by the end-to-end manifest. The source archive and manifest are identified by SHA-256 digests in the release notes. The archived CPC proofs retain their original two absolute signature include lines. The release verifier creates a temporary relocated copy by rewriting only those two lines, verifies that every remaining proof-body byte is unchanged, checks all 51 signature files against the official cvc5 1.3.3 source archive, and runs Ethos on the relocated copy. The full eight-cell relocation replay passed on the release host, preserving every proof-body byte; this is verifier-level relocatability, not an assumption about the archived paths or an independent-host replication.

Minimal verification. From the certification root:

```
shasum -a 256 -c EFX48_END_TO_END_MANIFEST.sha256
```

The release README gives platform-specific commands for the Lean build, semantic audit, formula-to-proof comparison, and Ethos checks.

12 Conclusion

Every nonnegative additive instance with four agents and eight goods admits a complete EFX allocation. The proof closes the first previously open four-agent cell beyond the $m \le n + 3$ existence frontier. It is a fixed-cell result, not a solution for four agents with arbitrary goods. Its mathematical reduction is formalized in Lean, while its terminal arithmetic is discharged by exact CPC proofs checked by Ethos under an explicit hybrid trust boundary.

The remaining mathematical challenge is explanatory rather than existential: give a human proof of Proposition 9.1, then use the same bundle geometry to compress the other seven cells. The remaining validation challenge is independent reproduction and external expert review.

A Terminal formula schema

For a fixed cell, the formula uses rational variables w_{ig} for $i \in \{0, 1, 2, 3\}$ and $g \in \{0, \dots, 7\}$.

Base valuation constraints.

$$w_{ig} > 0, \qquad \sum_{g=0}^{7} w_{ig} = 1, \qquad \bigvee_{g=0}^{7} (w_{ig} \neq w_{jg}) \quad (i \neq j).$$

Residual exclusions. For every good, at least one agent ranks another good below it, excluding a universal minimum. For every agent-good pair, the negation of the formal peelability conjunction is asserted.

Ranking cells. The favorite or favorite-and-second-favorite inequalities encode one row of Table 1. The strict ranking inequalities make the prescribed favorites unique.

Feasible edges. For each partition $\mathcal{B}$, observer i, and candidate own bundle B_b, an edge variable is definitionally equivalent to

$$\bigwedge_{\substack{c \neq b \\ g \in B_c}} \left(\sum_{h \in B_b} w_{ih} \ge \sum_{h \in B_c \setminus \{g\}} w_{ih} \right).$$

Matching obstruction. For every $\pi \in S_4$:

$$\bigvee_{i=0}^{3} \neg\, \mathrm{edge}(i, \pi(i)).$$

The conjunction over all 24 permutations says that the partition has no perfect matching.

B Published-input interface

The two external inputs are:

1. complete EFX for four agents and seven goods, obtained from Mahara's $m \leq n + 3$ theorem; and
2. complete EFX for three additive agents and seven goods, obtained from the three-agent theorem of Chaudhury, Garg, and Mehlhorn.

The pinned audits check exact PDF hashes, theorem fragments, completeness, and the EFX convention. They do not verify the papers' proofs. The Lean lifting transports the standard statements to the subtype-indexed deleted instances used by the residual cuts.

C Reviewer checklist

A mathematical reviewer should check:

1. the EFX convention and complete-allocation definition;
2. the uniform perturbation from nonnegative to positive weights;
3. row normalization and simultaneous generic perturbation;
4. the favorite-small-bundle theorem;
5. universal-minimum and peelable reinsertion;
6. canonical favorite and second-favorite coverage;
7. stabilizer soundness;
8. partition-to-perfect-matching equivalence; and
9. the exact two published theorem variants.

A formal/computational reviewer should check:

1. the Lean root imports and axiom audit;
2. the parser's complete SMT assertion coverage;
3. the manually reviewed Python-specification-to-Lean correspondence;
4. formula and proof hashes;
5. the 545,360-assumption comparison;
6. CPC signature relocation;
7. Ethos verdicts; and
8. replay on an independent machine.

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